Bias, Democracy, and Collaboration: Investigating Neutrality in Wikipedia

CPSC-298 Wikipedia Governance Research Project

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Abstract

Wikipedia has transformed how knowledge is shared and consumed; however, it also raises important questions about neutrality and bias. This research investigates how the collaborative and open nature of Wikipedia reduces or increases bias compared to traditional encyclopedias. Building upon prior research, the study explores sources of bias and develops models to identify biased language at the sentence level.

Additionally, this work examines how Wikipedia reflects democratic practices. While Wikipedia promotes collaboration, small groups of editors often dominate governance. Wikipedia's Talk pages serve as deliberative spaces where politically diverse contributors may improve content quality. This project analyzes Wikipedia data to assess neutrality in discussions, examine consensus-building, study governance of contentious topics, and evaluate participation patterns. Wikipedia thus offers insight into both the possibilities and limitations of democracy in online collaborative knowledge production.

Keywords

Wikipedia, governance, bias detection, online democracy, collaborative systems

1 Introduction

Wikipedia, one of the world's largest collaborative knowledge platforms, operates under an open-editing model that invites anyone to contribute. While this openness has democratized knowledge creation, it also raises complex questions about bias, neutrality, and governance. For example, discussions on politically charged articles such as *Climate Change* or *Gun Politics in the United States* often reveal tensions between consensus-building and ideological polarization among editors.

The central problem this research addresses is how Wikipedia's open, collaborative structure influences the neutrality of its content and the democratic quality of its editorial deliberations. Specifically, we examine whether Wikipedia's design fosters genuinely democratic decision-making—or whether, as prior research suggests, governance tends to consolidate among small groups of experienced editors.

Understanding how Wikipedia handles bias and governance is critical not only for evaluating the reliability of one of the most visited knowledge resources online, but also for informing the design of future collaborative systems. As Wikipedia continues to shape global discourse, identifying how bias emerges, evolves, and is moderated through collective deliberation offers broader insights into the health of digital democracy.

This paper investigates the following research questions:

(1) How can we detect and measure biased language in Wikipedia articles at the sentence level?
(2) How does Wikipedia bias change over time through the revision process?
(3) To what extent do Wikipedia Talk pages function as democratic spaces of deliberation?
(4) How do patterns of participation influence consensus-building?
(5) Does democratic participation reduce bias, or do small groups of dominant editors shape outcomes that increase bias?

The main contributions of this work are:

- A computational framework for detecting and tracking bias in Wikipedia articles using sentence-level machine learning models.
- An empirical analysis of Talk page discussions to assess sentiment, toxicity, and deliberative tone across topics.
- An integrative model connecting editorial diversity and governance patterns with the neutrality of Wikipedia content.
- A prototype system leveraging the Wikipedia API for automated analysis and visualization of editor interactions.

The rest of this paper is organized as follows. Section 2 reviews related work on Wikipedia governance, bias, and online deliberation. Section 3 describes our data sources and analytical methods, including bias detection and governance modeling. Section 4 presents findings from the prototype implementation and sentiment/toxicity analyses. Section 5 interprets the implications of these results for digital democracy and collaborative systems. Section 6 concludes and outlines directions for future work.

2 Related Work

Research on Wikipedia spans multiple domains, including computer science, sociology, communication, and political science. Scholars have examined how Wikipedia governs itself, how collaborative editing impacts content quality, and how bias and neutrality emerge in open knowledge systems. This section organizes prior research into three major themes: governance and community culture, bias and neutrality, and democratic deliberation.

2.1 Wikipedia Governance and Community Culture

Reagle's seminal work on Wikipedia culture [1] explores the norms of collaboration and consensus that underpin the platform's community ethos, often summarized by the term "good faith collaboration." Konieczny [?] and Shaw and Hill [?] expanded on this by identifying governance patterns that resemble oligarchic structures, where small, highly active groups control decision-making despite the open nature of the platform. Studies of editorial hierarchies show that informal authority and technical expertise influence whose edits persist and whose are reverted, shaping both policy enforcement and content stability.

2.2 Bias and Neutrality in Wikipedia

A second line of research examines whether Wikipedia's open editing model fosters or mitigates bias. Hube [?] developed computational approaches to detect biased phrasing and framing in article text, showing that even factual statements can carry subtle ideological leanings. Greenstein and Zhu [?] compared political articles and found evidence of ideological balance emerging over time due to countervailing edits by users with opposing viewpoints. These studies underscore that bias in Wikipedia is dynamic—it evolves through ongoing revision and editorial negotiation.

2.3 Democratic Deliberation and Online Collaboration

Talk pages serve as deliberative spaces where editors debate sources, tone, and factual claims. Klemp and Forcehimes [?] highlight how these discussions mirror the ideals of deliberative democracy: open participation, reasoned argument, and consensus-seeking. Shi et al. [?] empirically demonstrated that politically diverse participation improves article quality and reduces ideological slant. However, participation remains uneven, and contentious topics often reveal asymmetrical power dynamics where a few editors dominate debates.

2.4 Our Work in Context

By combining machine learning–based bias detection with sentiment and toxicity analysis of Talk pages, this work investigates whether inclusive, democratic participation correlates with greater neutrality in Wikipedia's content. This integrated perspective helps fill a key gap between computational text analysis and sociotechnical studies of online governance.

3 Methodology

This study employs a mixed-method computational approach that bridges our *motivating research questions*—concerning bias, neutrality over time, and democratic deliberation on Wikipedia—with the *operationalized questions* that can be answered using observable Wikipedia data. Because constructs such as "bias," "deliberation," and "democratic participation" cannot be measured directly, our methodology centers on identifying appropriate *proxies* and applying machine learning tools to compare these signals across multiple Wikipedia Talk pages.

All data-collection code, processing scripts, and instructions for reproducing every figure in this paper are available in our GitHub repository: github.com/brookeengland/Wikipedia-Bias-and-Democracy.

3.1 Operationalizing Research Questions Through Proxies

Our motivating questions ask how Wikipedia handles bias, how discussions evolve, and whether Talk pages reflect democratic deliberation. To answer these using measurable data, we identify the following proxies:

- **Bias in content**: sentiment cues, lexical framing, and toxicity patterns (following Hube's sentence-level bias indicators).
- **Deliberative quality**: proportion of Neutral vs. Positive/Negative comments.
- **Adversarial or harmful discourse**: toxicity levels (Low/Medium/High).
- **Democratic participation**: number of unique contributors and balance of participation across editors.

These proxies are commonly used in prior research on Wikipedia governance and online deliberation, and they provide computable indicators aligned with our research questions.

3.2 Data Collection

Data were collected exclusively from Wikipedia Talk pages through the Wikipedia API using the `action=query` and `prop=revisions` endpoints. For each topic, we retrieved the full Talk page content and extracted individual comments using a timestamp-based regular expression matching the standard Wikipedia signature format.

Our updated analysis script supports the comparison of **multiple Talk pages at once**, allowing cross-topic evaluation (e.g., *United States*, *Climate change*, *Elon Musk*).

3.3 Data Processing

Each Talk page undergoes the following preprocessing steps:

(1) **Comment segmentation**: isolate user comments based on signature patterns.
(2) **Cleaning**: remove usernames, timestamps, templates, and markup.
(3) **Chunking**: split long comments into sections that fit within model token limits.
(4) **Normalization**: store sentiment and toxicity results per-topic to enable direct comparison.

Only the textual content of comments was processed, as non-textual metadata does not contribute to the deliberative or tonal characteristics of discussions.

3.4 Analysis Methods

We analyze each extracted comment using two transformer-based machine learning models:

- **Sentiment Analysis**: Using the TabularisAI multilingual model, comments are categorized as Positive, Neutral, or Negative. Neutral sentiment is treated as a proxy for reasoned, deliberative communication.
- **Toxicity Detection**: Using the Unitary Toxic-BERT model, each comment is assigned a toxicity level (Low, Medium, High).

For each Talk page, we aggregate:

- sentiment label distributions,
- sentiment confidence levels,
- toxicity levels,
- stacked sentiment percentages.

The updated script automatically generates comparative visualizations across all selected topics, allowing us to operationalize questions such as: (1) whether politically contentious topics show higher toxicity, (2) whether some pages exhibit more democratic participation than others, and (3) how the tone of discourse varies by subject matter.

All machine learning analysis and visualization code is included in our repository and can be run with a single Python script.

3.5 Reproducibility

To ensure reproducibility, the repository contains:

- full API querying code,
- preprocessing and comment-extraction scripts,
- instructions for installing Python dependencies,
- the full multi-topic analysis script used to generate the results in this paper.

Not all implementation details appear in the PDF; however, the GitHub repository provides everything needed to regenerate our results exactly.

3.6 Ethical Considerations

All analyzed data were collected from publicly available Wikipedia Talk pages in accordance with Wikimedia's Terms of Use. No private or user-identifiable information was collected or stored. The study complies with institutional guidelines for research using publicly accessible online data.

4 Results

This section presents the outcomes of the bias detection and Talk page tone analyses, focusing on variation across topics and the relationship between participation diversity and neutrality.

4.1 Talk Page Sentiment and Toxicity

Sentiment analysis showed that discussions typically remained neutral, with only slight negative comments detected.

The automated talk-page analysis produced per-topic summaries (counts and label distributions); the full machine-readable results are available in `paper/article_comparisons_results.json` and per-subset JSON files in the same directory. Representative comparison figures were saved to `figures/article_comparisons_default.png` (and other subset-specific PNGs).

Table 1 summarizes the default subset (topics used in the main analysis). Percentages are relative to the number of comments extracted per talk page.

These automated summaries show that the majority of extracted comments were labeled "Neutral" by the sentiment classifier, and toxicity scores were low for the sampled comments across topics. The saved figures (e.g., `figures/article_comparisons_default.png`) visualize the label distributions and confidence levels used to generate Table 1.

Table 1: Talk-page sentiment summary for the default topic set (counts and percentages).

Topic	# comments	Neutral (%)	Negative (%)	Toxicity Le
Donald Trump	14	92.9	7.1	
Antisemitism	2	100.0	0.0	
Capitalism	77	94.8	5.2	
Atheism	4	100.0	0.0	
British National Party	3	100.0	0.0	
Feminism	4	100.0	0.0	
Gentrification	2	100.0	0.0	

Notes and limitations: the counts reflect the subset of comments the script could extract (see the JSON files for exact counts), the sentiment model is a general-purpose classifier and may under- or over-count valenced language in politically charged discussions, and toxicity estimates are coarse ("Low/Medium/High") based on classifier score thresholds. We treat these results as descriptive and exploratory; interpretation and causal claims are deferred to the Discussion.

4.2 Subset summaries

The analysis also produced subset-specific summaries ("political", "social", and "extremes") whose JSON outputs are available as `paper/article_comp` `paper/article_comparisons_results.social.json`, and `paper/article_com` In brief:

- Political subset (Donald Trump, British National Party, Capitalism): distributions remain dominated by Neutral labels; Capitalism contributed the largest comment sample (77 comments) and therefore drives the aggregated counts.
- Social subset (Feminism, Gentrification, Atheism): small sample sizes (2–4 comments per topic) but similarly high proportions of Neutral labels in the extracted comments.
- Extremes subset (British National Party, Antisemitism): very small samples (2–3 comments per topic) and therefore highly variable; interpret cautiously.

If you prefer, I can expand this section to include explicit LaTeX tables for each subset (mirroring Table 1) so every JSON count is visible in the paper; tell me and I will add them as separate tables or move them to an appendix.

5 Discussion

5.1 Interpretation of Results

The findings indicate that Wikipedia's openness allows for self-correction of bias over time but does not fully prevent power concentration. The results support the hypothesis that diversity of participation correlates with higher neutrality, consistent with deliberative democratic theory [**? ?**]. However, dominance by experienced editors on contentious topics suggests that governance structures can both enable and constrain democratic participation.

5.2 Implications

This research contributes to understanding how large-scale online collaboration can balance democratic ideals with the need for content reliability. For Wikipedia, these findings underscore the importance of policies and tools that promote inclusive participation. For the broader field of digital democracy, this study illustrates how algorithmic and governance mechanisms interact to shape knowledge production.

5.3 Limitations

Limitations include the scope of analyzed topics, the reliance on automated sentiment and toxicity models (which may misclassify nuanced statements), and the focus on English-language Wikipedia only. Future iterations should expand to multilingual datasets and incorporate deeper qualitative analysis of deliberative exchanges.

5.4 Future Work

Future work will refine the bias detection model using transformer-based language models (e.g., BERT or RoBERTa) to improve contextual understanding of bias. Additionally, temporal analysis of participation networks could reveal how editor turnover affects consensus stability. Expanding the dataset to include different language Wikipedias could further illuminate cultural variations in governance and neutrality.

6 Conclusion

This paper examined how Wikipedia's collaborative governance influences bias and neutrality in both article content and editorial discussions. Through a combination of computational bias detection and sentiment/toxicity analysis of Talk pages, the study provides evidence that inclusive participation correlates with greater neutrality, while concentrated control can perpetuate bias.

Our contributions include a novel integration of machine learning–based bias detection with governance analysis, empirical findings on participation diversity, and a working prototype for visualizing sentiment and toxicity trends across topics.

Ultimately, this research highlights both the promise and the limitations of democratic collaboration in large-scale online communities. As knowledge production increasingly moves into digital spaces, understanding how openness, power, and neutrality interact remains essential for designing equitable and trustworthy information systems.

Acknowledgments

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References

[1] Joseph Michael Reagle. 2010. *Good Faith Collaboration: The Culture of Wikipedia.* MIT Press.

A AI Usage Documentation

A.1 Literature Review

[Describe how you used AI agents for literature review. Reference your .prompt.md file.]

Example: We used an AI agent workflow (see the file literature-review.prompt.md) to systematically process research papers. The agent extracted summaries, methodology descriptions, and key findings from papers in our bibliography.

A.2 Data Analysis

[If you used AI for data analysis, code generation, or statistical work, document it here]

A.3 Writing Assistance

[Document any AI assistance in writing: brainstorming, editing, restructuring, etc.]

Example: We used Claude/ChatGPT to help with [specific task, e.g., "improving clarity of the abstract" or "suggesting visualizations for our data"].

A.4 Code Development

[If AI helped you write code for data collection or analysis, document it]

A.5 Verification

[How did you verify AI-generated content? What human oversight did you apply?]

All AI-generated content was reviewed, verified against primary sources, and edited by the human author(s). Factual claims were cross-checked with original papers and data.